

Mechanical Engineering Learning Tutorial

Linear Regression and Logistic Regression

A beginner-friendly tutorial for Mechanical Engineering students: from intuition to mathematics to engineering interpretation

Prepared as a self-study and teaching resource. The tutorial opens with a detailed outline and then develops the topic from first principles without any programming content.

Audience	Mechanical Engineering students who are new to artificial intelligence and machine learning
Background assumed	Basic algebra, calculus, and introductory linear algebra; no prior knowledge of data-driven modelling is required
Focus	Intuition, mathematical mounting bases, model interpretation, evaluation, and Mechanical Engineering applications
Deliberate exclusions	Programming, coding, software implementation, and platform-specific instructions

Prepared for introductory study in data-driven modelling for Mechanical Engineering

Detailed Outline and Learning Roadmap

This tutorial is intentionally organised as a progression from intuition to mathematical formulation, then to interpretation, evaluation, and Mechanical Engineering use. Students should understand each stage conceptually before moving to the next one. The outline below serves as a roadmap for reading and revision.

1. 1. Mounting bases of artificial intelligence and machine learning
 - The tutorial begins by defining artificial intelligence (AI), machine learning (ML), supervised learning, regression, and classification. It explains why these ideas matter in Mechanical Engineering and where Linear Regression and Logistic Regression fit within the broader landscape of data-driven modelling.
2. 2. Intuition behind regression-based models
 - Before formal equations appear, the tutorial explains how a model uses measured input variables to estimate an output. It introduces observations, predictor variables, target variables, and the difference between predicting a numerical quantity and predicting a class or category.
3. 3. Linear Regression concepts
 - The next stage introduces Linear Regression as a model for continuous target variables. It explains simple linear regression, multiple linear regression, fitted lines, fitted hyperworkcells, residuals, coefficients, intercepts, and the practical assumptions that make the model useful.
4. 4. Mathematical mounting bases of Linear Regression
 - After intuition has been established, the document presents the Linear Regression equations in scalar and matrix form. It explains ordinary least squares (OLS), fitted values, residuals, the sum of squared errors, and the logic of minimisation.
5. 5. Interpretation and evaluation of Linear Regression
 - The tutorial then shows how to interpret coefficients in engineering units, how to judge goodness of fit, and how residual analysis and common error measures help determine whether a fitted model is credible.
6. 6. Logistic Regression concepts
 - The document next turns to Logistic Regression, explaining why it is used for classification despite its name. It introduces binary classification, probabilities, odds, log-odds, the logistic or sigmoid function, class membership probabilities, and decision thresholds.
7. 7. Mathematical mounting bases of Logistic Regression
 - Once the classification intuition is clear, the tutorial shows the logistic transformation mathematically and explains the likelihood idea, maximum likelihood estimation (MLE), and the loss function used to estimate coefficients.
8. 8. Interpretation and evaluation of Logistic Regression
 - The guide then explains how to interpret predicted probabilities and odds ratios, how classification decisions are made, and how measures such as the confusion matrix, accuracy, precision, and recall should be interpreted in an engineering context.
9. 9. Comparison of the two models
 - A direct comparison highlights what the models share and where they differ. In particular, the tutorial emphasises the distinction between fitting a line to a continuous quantity and estimating the probability that an observation belongs to a class.
10. 10. Mechanical Engineering applications
 - Several applications are presented from materials, structures, machined surfaces, thermal-fluid systems, and mechanics of materials. These examples connect the mathematical models to realistic Mechanical Engineering questions and show why interpretable statistical learning methods remain valuable.
11. 11. Worked examples

- A simplified Linear Regression example and a simplified Logistic Regression example are developed in detail. Each one focuses on engineering meaning, not software implementation, so that the relation between model form and engineering judgement remains clear.
12. 12. Assumptions, strengths, and limitations
 - The document then addresses when these models perform well, when they struggle, and what assumptions or data issues can cause misleading results. Topics include nonlinearity, outliers, multicollinearity, and class imbalance.
 13. 13. Beginner support and study guidance
 - The final substantive section explains training and testing, overfitting, underfitting, and generalisation. It also lists common mistakes beginners make and offers practical advice on deciding whether an engineering problem is a regression problem or a classification problem.
 14. 14. Conclusion and reference framework
 - The tutorial closes by recapping the main ideas and reinforcing the conceptual distinction between Linear Regression and Logistic Regression. A full reference list is included so that students can continue studying from academically credible sources.

Table 1. Recommended study production line through the tutorial.

Stage	Main question	Expected understanding before moving on
Mounting bases	What kinds of problems do AI and ML solve?	You can distinguish supervised learning, regression, and classification.
Intuition	What do inputs and outputs mean?	You can identify predictors, targets, and the role of data in modelling.
Linear Regression	How is a continuous quantity modelled?	You understand lines, hyperworkcells, coefficients, residuals, and least squares.
Logistic Regression	How is a class probability modelled?	You understand probabilities, odds, log-odds, and decision thresholds.
Interpretation and applications	How should results be used in engineering?	You can interpret coefficients, evaluate performance, and connect model outputs to Mechanical Engineering decisions.

Part I. Mounting bases

From general ideas in artificial intelligence and machine learning to the basic intuition behind regression and classification.

1 Introduction to Artificial Intelligence and Machine Learning

Artificial intelligence (AI) is the broad field concerned with building systems that can perform tasks associated with human reasoning, pattern recognition, prediction, or decision-making. Machine learning (ML) is a major subfield of AI in which mathematical models learn useful patterns from data rather than being programmed with every rule explicitly. In engineering language, machine learning offers a systematic way to use historical observations, measurements, or test results to support prediction and classification tasks [2,5].

A central idea in ML is supervised learning. In supervised learning, the modeller has examples for which both the input variables and the correct output are known. The model studies these labelled examples and learns a relationship that can later be used on new observations. If a Mechanical Engineering laboratory has past measurements of material ingredients and the corresponding compressive strengths, those paired data can be used for supervised learning [1,2].

Two of the most important supervised learning problem types are regression and classification. Regression means predicting a numerical quantity on a continuous scale, such as deflection in millimetres, deflection in centimetres, or compressive strength in megapascals. Classification means assigning an observation to a class or category, such as stable or unstable slope, compliant or non-compliant coolant sample, or distressed and non-distressed machined surface section [1,2,4].

Linear Regression and Logistic Regression are among the most widely taught introductory models because they are mathematically transparent, interpretable, and foundational. Linear Regression addresses continuous prediction problems. Logistic Regression addresses classification problems by modelling the probability that an observation belongs to a class. Although more complex ML methods exist, these two models remain essential because they clarify core ideas that reappear throughout statistical learning and engineering data analysis [1-4].

For Mechanical Engineering students, these methods are relevant because many engineering questions involve measurable inputs and meaningful outputs. Materials engineers relate mix composition to strength. Structural engineers relate load, geometry, and stiffness to deflection. Mechanics of Materials and Structures engineers estimate deflection or failure likelihood from material and loading conditions. Water and environmental engineers assess the chance that a sample exceeds a regulatory threshold. In all these cases, data-driven models can complement physical understanding and support engineering judgement [5-10].

Table 2. Core introductory terms used throughout the tutorial.

Term	Plain-language meaning	Mechanical Engineering example
Artificial intelligence	A broad field concerned with intelligent or adaptive behaviour in machines	Automated support for mechanical systems inspection or maintenance prioritisation
Machine learning	Methods that learn useful patterns from data	Learning from historical laboratory or test measurements
Supervised learning	Learning from examples with known outputs	Using measured inputs and known

		test results to build a predictive model
Regression	Predicting a numerical quantity	Predicting deflection, strength, or deflection
Classification	Predicting a category or class	Classifying material as suitable or unsuitable

2 Intuition Behind Regression-Based Models

Every supervised learning model begins with observations. An observation is one case, record, or measured instance. For a material dataset, one observation might describe a single test coupon location. For a materials dataset, one observation might describe one material sample and its measured strength. Each observation contains input variables, also called predictor variables, and an output variable, also called the target or response variable [1,3].

The intuitive goal is simple: use measured inputs to say something useful about an output that matters. If the relationship between inputs and outputs is systematic rather than entirely random, a model can capture part of that pattern. The model does not claim that reality is perfectly determined by the inputs. Instead, it provides an approximate rule that links the information we have to the output we want to estimate. The unexplained part remains as error or uncertainty.

It is helpful to distinguish two different kinds of outputs. A continuous output can take many numerical values on a scale, such as 31.6 millimetres of deflection or 47.5 megapascals of strength. A categorical output belongs to a class, such as yes or no, acceptable or unacceptable, or low risk and high risk. This single distinction usually determines whether a beginner should think first about regression or classification.

Regression-based models are therefore not defined merely by the fact that they contain equations. They are defined by the kind of question they answer. If the question is “how much?”, a regression model is needed. If the question is “which class?” or “with what probability does a class occur?”, a classification model is needed. Linear Regression answers the first type. Logistic Regression answers the second type [1,4].

Before any equation is written, students should form a visual and conceptual picture. In Linear Regression, one imagines a line or flat surface passing through scattered data so that the predicted value changes smoothly with the inputs. In Logistic Regression, one imagines a continuous score being converted into a probability between 0 and 1, after which a decision threshold turns that probability into a class label. This mental picture makes the mathematics much easier to understand later.

Table 3. Deciding whether a Mechanical Engineering problem is a regression problem or a classification problem.

Engineering question	Output type	Model family	Typical output
Predict 28-day mechanical material strength from mix properties	Continuous	Regression	A number in MPa
Predict immediate deflection beneath a footing	Continuous	Regression	A number in mm or cm
Classify a material sample as suitable or unsuitable for a given use	Categorical	Classification	A class label
Predict whether machined-surface defect will	Categorical	Classification	Yes or No

occur within the next year			
Estimate beam deflection under service load	Continuous	Regression	A numerical deflection
Predict whether coolant quality will comply with a standard	Categorical	Classification	Compliant or non-compliant

Part summary

- AI is the broader field; ML is the part that learns patterns from data.
- Supervised learning uses examples with known outputs to learn a predictive relationship.
- The first major decision for a beginner is whether the target is continuous or categorical; this usually separates regression from classification.

Part II. Linear Regression

Understanding continuous prediction, least squares, coefficients, residuals, and engineering interpretation.

3 Linear Regression: Concepts and Engineering Meaning

Linear Regression is used when the quantity to be predicted is numerical and continuous. The model expresses the predicted output as a linear combination of the input variables. The word linear refers to linearity in the coefficients: each predictor contributes through a coefficient that scales its effect on the prediction [1,3].

In statistics, the output is often called the dependent variable or the response variable. In ML, the same quantity is often called the target variable. The inputs are called independent variables in classical statistics and predictors or features in ML. These are different terminologies for closely related ideas, and beginners should become comfortable moving between them.

Table 4. Common vocabulary used for inputs and outputs in Linear Regression.

Statistics term	Common ML term	Meaning
Independent variable	Predictor or feature	An input used to help explain or predict the output
Dependent variable	Target or response	The output that the model is trying to predict
Observation	Sample or case	One row of data describing one measured instance
Error term	Noise or unexplained variation	The part of the outcome not captured by the model

Simple linear regression uses one predictor. If material strength is predicted using only water-cement ratio, the fitted model can be represented as a straight line on a two-dimensional graph. Multiple linear regression uses several predictors at once. If strength is predicted from water-cement ratio, curing age, cement content, and aggregate properties together, the model becomes a flat surface in a higher-dimensional space. This higher-dimensional analogue of a line is called a hyperpworkcell.

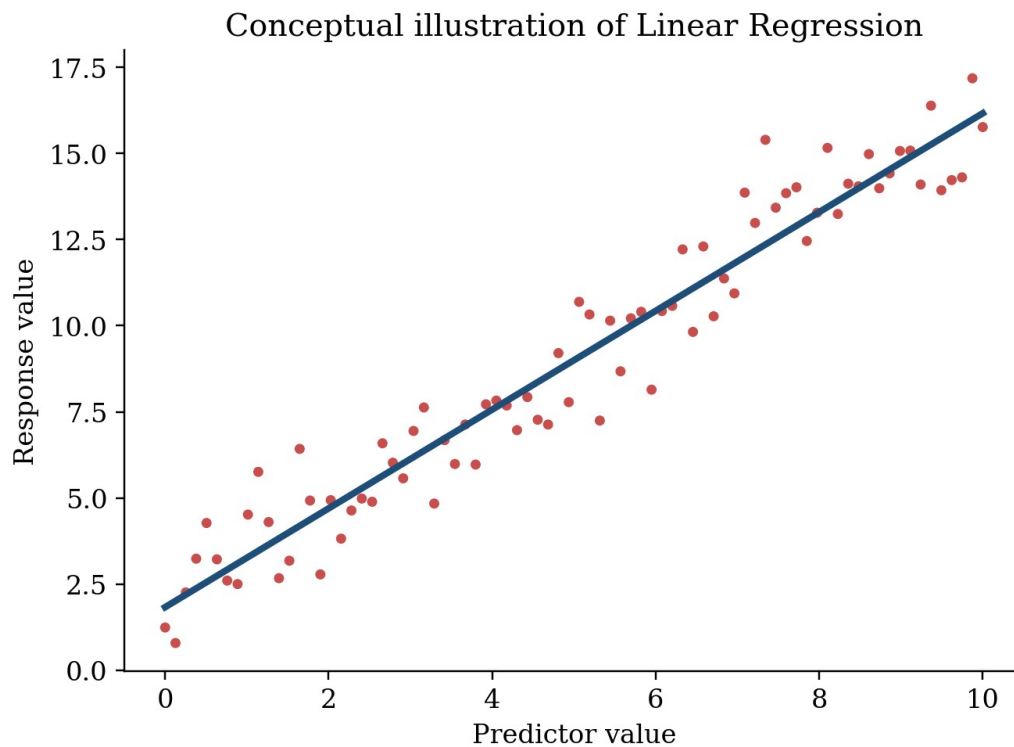


Figure 1. Conceptual illustration of a fitted straight line passing through scattered observations. Adapted from Berland (2007) [11].

The fitted line or fitted hyperpworkcell is the model's best estimate of the systematic trend in the data. For each observation, the model produces a fitted value, meaning the value predicted by the fitted equation. The difference between the observed value and the fitted value is called the residual. Residuals are central because they reveal what the model does not explain.

Coefficients describe how the predicted output changes when predictors change. In a simple linear model, the slope coefficient tells us how much the predicted response changes for a one-unit increase in the predictor. In a multiple regression model, a coefficient tells us how much the prediction changes when one predictor increases by one unit while the other predictors are held fixed. The intercept is the predicted output when all predictors are zero. Sometimes that value is physically meaningful, and sometimes it is only a mathematical anchor point.

For engineering interpretation, units matter. A coefficient has units inherited from the output and predictor. If deflection is measured in millimetres and depth is measured in metres, the corresponding slope coefficient has units of millimetres per metre. This unit-based interpretation is one reason Linear Regression remains especially useful in engineering education and professional communication [3].

Linear Regression is built on several assumptions. These assumptions do not mean that every real dataset must be perfect. Rather, they identify the conditions under which the model's interpretation and statistical properties are most reliable.

Table 5. Practical assumptions of Linear Regression explained in accessible language.

Assumption	Plain-language meaning	Why it matters
Linearity	The average response changes approximately linearly with the predictors	A straight-line form should be a reasonable first approximation
Independence	Errors from one observation should not systematically depend on errors from another	Dependence can make estimated uncertainty misleading

Constant variance	Residual spread should stay roughly similar across the predictor range	Large changes in spread can distort inference and prediction quality
Approximately normal errors	Residuals should be roughly bell-shaped if formal inference is needed	This matters especially for confidence intervals and significance tests
No severe multicollinearity	Predictors should not be almost exact duplicates of each other	Severe redundancy makes coefficients unstable and hard to interpret

4 Mathematical Mounting bases of Linear Regression

To present the mathematics clearly, let n denote the number of observations. For observation i , let x_i be the predictor value and y_i be the observed response value. The coefficient β_0 represents the intercept, β_1 represents the slope, and ϵ_i represents the part of the observation not explained by the linear relation. With this notation, the simple linear model can be written as follows [1,3].

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$$e_i = y_i - \hat{y}_i$$

The fitted quantities use hats, which indicate estimated rather than unknown population parameters. Thus $\hat{y}_i$ is the predicted value produced by the fitted model, and e_i is the residual. Residuals are observable after fitting because they compare the actual response with the fitted response for each observation.

The ordinary least squares (OLS) principle chooses the coefficients that make the residuals collectively as small as possible. Specifically, it minimises the sum of squared residuals. Squaring serves three purposes. First, it prevents positive and negative residuals from cancelling. Second, it penalises large errors more strongly than small ones. Third, it yields a mathematically convenient optimisation problem with a closed-form solution in the linear case [1,3].

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

For students familiar with calculus, the coefficients above arise by differentiating the sum of squared errors with respect to β_0 and β_1 and then setting the derivatives equal to zero. This produces

the normal equations, whose solution gives the least-squares line. The fitted line is therefore not arbitrary; it is the line that minimises a precise objective function.

Multiple linear regression extends the same idea to several predictors. If each observation has p predictors, then the model uses coefficients β_1 through β_p to describe how each predictor contributes to the prediction. In matrix form, the entire dataset can be written compactly as follows [1,3].

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Here $\mathbf{y}$ is the n -by-1 vector of observed responses, $\mathbf{X}$ is the design matrix containing the predictors, $\boldsymbol{\beta}$ is the vector of unknown coefficients, and $\boldsymbol{\varepsilon}$ is the vector of errors. The matrix formula for $\hat{\boldsymbol{\beta}}$ is valid when $\mathbf{X}$ -transpose $\mathbf{X}$ is invertible. From a linear algebra viewpoint, OLS projects the observed response vector onto the column space of $\mathbf{X}$. Geometrically, the fitted values are the closest point in that subspace to the observed response vector under Euclidean distance.

This matrix view matters because it reveals the common structure shared by simple and multiple regression. The model is still linear in the coefficients, and the fitted solution is still found by minimising squared residuals. The only difference is that the fitted object is no longer a line in two dimensions but a hyperplane in a higher-dimensional predictor space.

5 Model Interpretation and Evaluation for Linear Regression

A Linear Regression coefficient should always be interpreted in engineering units. Suppose the coefficient of beam span is positive. Then, holding other predictors fixed, increasing span raises the predicted response by the coefficient amount per unit increase in span. If the response is deflection, that coefficient should be discussed in terms of how many millimetres of additional deflection are associated with one additional metre of span. This unit-based interpretation is much more informative than merely saying that the coefficient is positive or negative.

Goodness of fit describes how well the fitted model captures variation in the observed data. One common measure is the coefficient of determination, written as R -squared. It compares the residual sum of squares with the total variation in the response. Values closer to 1 indicate that the model explains a larger proportion of the variation in the observed data, but a high R -squared alone does not prove that the model is appropriate or physically meaningful [1,3].

$$R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE) summarise prediction error in different ways. MAE gives the typical absolute error and is easy to interpret in the units of the response. MSE penalises large errors strongly because of the square. RMSE returns to the original units of the response while still reflecting the squared-error penalty. In engineering practice, the preferred metric often depends on whether occasional large errors are especially costly.

Residual analysis is equally important. A residual plot places residuals on the vertical axis and fitted values or predictor values on the horizontal axis. If the model is appropriate, one hopes to see random scatter around zero rather than a systematic pattern. Curvature can indicate nonlinearity. A fan shape can indicate non-constant variance. Unusually extreme residuals can suggest outliers or data problems.

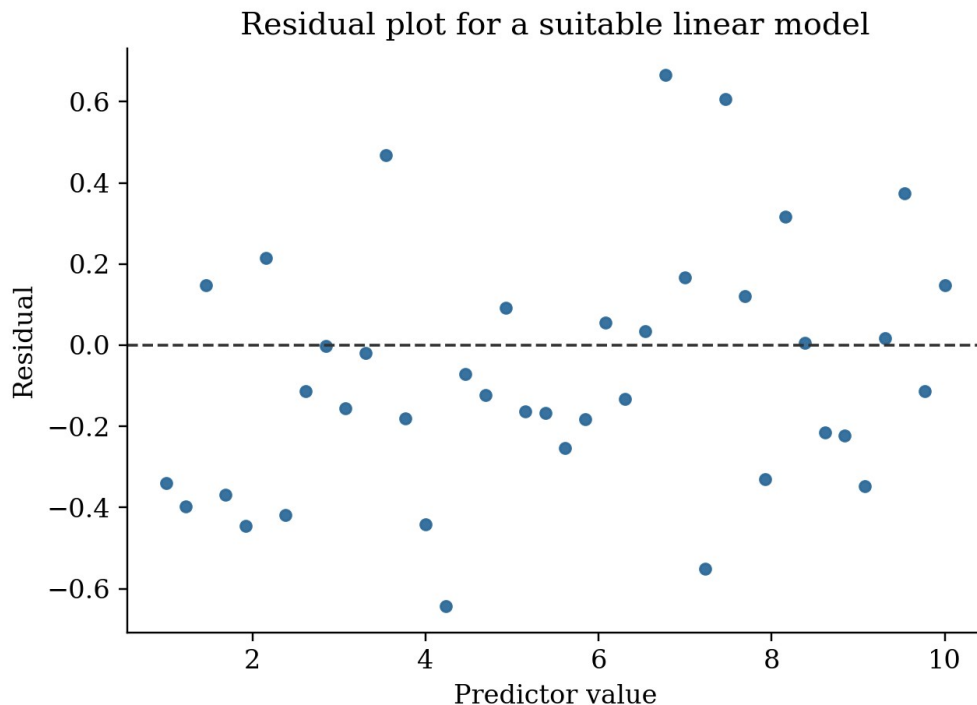


Figure 2. Residuals scattered randomly around zero suggest that a linear model may be a reasonable first approximation. Adapted from ShristiV (2019) [12].

From an engineering perspective, model interpretation should never stop with numerical fit statistics. A coefficient should have a plausible sign. A predicted quantity should stay within a physically sensible

range. Extrapolation beyond the observed data range should be treated cautiously. A model that fits past data well but contradicts basic mechanics, materials knowledge, or mounting conditions should not be accepted uncritically.

Part summary

- Linear Regression predicts a continuous target by fitting a line or hyperpworkcell to data.
- OLS estimates the coefficients by minimising the sum of squared residuals.
- Interpretation requires attention to units, assumptions, residual behaviour, and engineering plausibility, not only to R-squared.

Part III. Logistic Regression

Understanding classification, probabilities, odds, log-odds, likelihood, and threshold-based decisions.

6 Logistic Regression: Concepts and Engineering Meaning

Logistic Regression is used when the target variable is categorical, especially when there are two possible outcomes. A common beginner example is yes versus no, but in Mechanical Engineering the same structure appears as stable versus unstable, compliant versus non-compliant, distressed versus non-distressed, or suitable versus unsuitable [1,4].

The name can initially be confusing. Logistic Regression is used for classification, not for continuous prediction. It is called a regression model because it still builds a linear predictor from the input variables. The key difference is that this linear predictor is not used directly as the final output. Instead, it is transformed into a probability between 0 and 1 by the logistic function [1,2,4].

In binary classification, the two classes are usually encoded as 1 and 0. For instance, $y = 1$ may mean that machined-surface defect occurs, while $y = 0$ means that it does not occur. The fitted model estimates the probability that $y = 1$ for a given set of predictor values. The model output is therefore not merely a label; it is a probability of class membership.

Probability measures how likely an event is on a scale from 0 to 1. Odds express the same idea differently. If the probability of an event is p , then the odds are p divided by 1 minus p . Odds equal 1 when an event is equally likely to occur or not occur. Odds exceed 1 when occurrence is more likely than non-occurrence, and odds are below 1 when occurrence is less likely. Logistic Regression often becomes easier to interpret once students understand this probability-odds relationship [4].

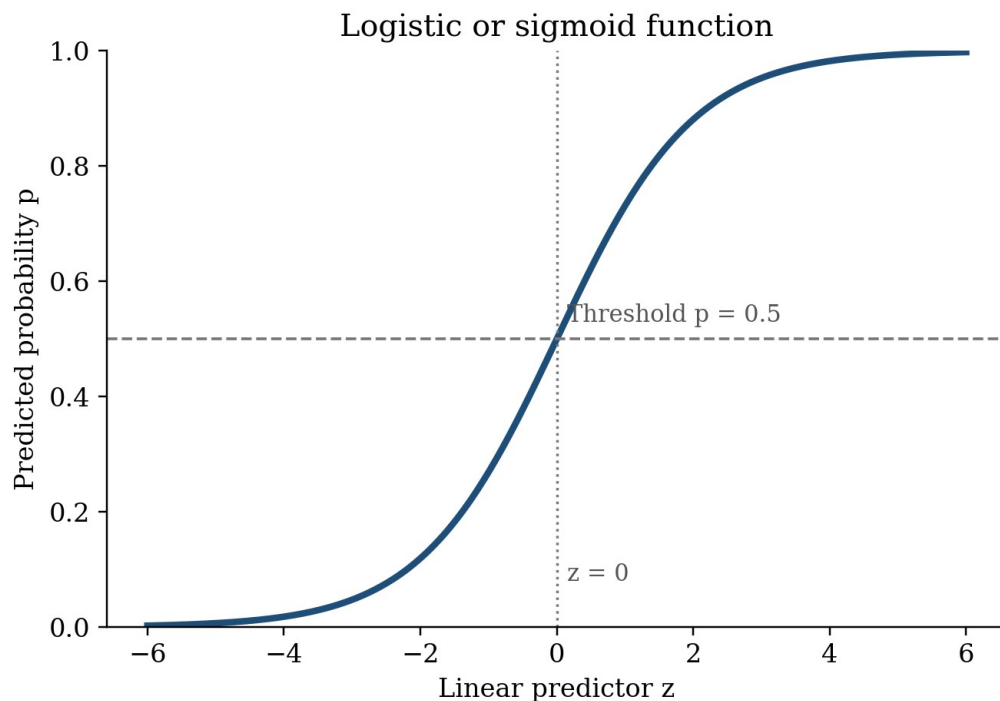


Figure 3. The logistic or sigmoid function transforms a linear predictor into a probability between 0 and 1; the horizontal line at 0.5 illustrates a common decision threshold. Adapted from Canley (2022) [13].

The S-shaped logistic curve is sometimes called the sigmoid curve. When the linear predictor is very negative, the predicted probability is close to 0. When the linear predictor is very positive, the probability is close to 1. Around the middle, small changes in the linear predictor can have a noticeable

effect on probability. This behaviour is well suited to classification problems because the model never produces impossible probabilities below 0 or above 1 [1,4].

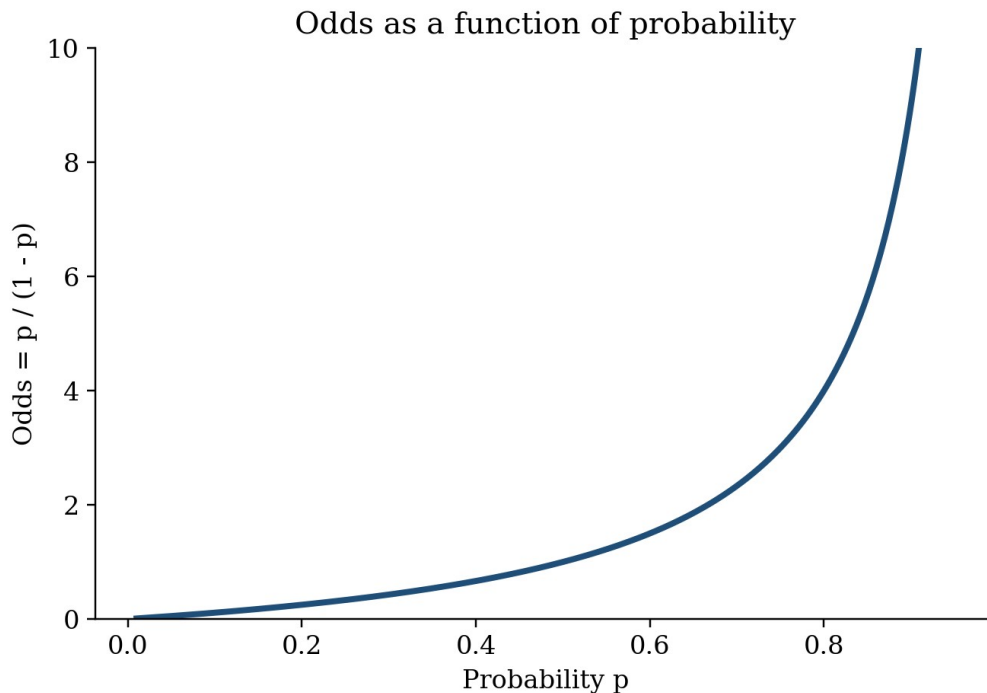


Figure 4. Odds increase nonlinearly as probability approaches 1. Adapted from ARAKI Satoru (2020) [14].

A classification decision is obtained by comparing the predicted probability with a decision threshold. A threshold of 0.5 is common in introductory explanations, but it is not mandatory. In a high-consequence engineering problem, such as bearing failure warning, one may prefer a lower threshold in order to reduce the chance of missing a dangerous case. Thus Logistic Regression naturally separates probability estimation from decision-making, which is an important conceptual advantage.

The coefficients in Logistic Regression do not act directly on the probability scale. Instead, each coefficient changes the log-odds of the target class. If a coefficient is positive, increasing that predictor raises the log-odds and therefore tends to increase the probability of class 1. If a coefficient is negative, increasing the predictor reduces the log-odds and therefore tends to decrease the probability [4].

Logistic Regression can also be extended beyond two classes. One common extension is multinomial logistic regression, which handles several unordered categories. Another extension is ordinal logistic regression, which is especially relevant when the categories have a natural order, such as poor, fair, and good structural condition. For beginners, however, binary Logistic Regression is the best place to master the core ideas [4,8].

7 Mathematical Mounting bases of Logistic Regression

Let y_i denote the class label for observation i , where y_i equals 1 if the target class occurs and 0 otherwise. As in Linear Regression, we first build a linear predictor from the inputs. The crucial difference is that this linear predictor, denoted z_i , is passed through the logistic function to obtain a probability p_i [1,4].

$$z_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}$$

$$p_i = P(Y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + e^{-z_i}}$$

This formula shows the close structural relation between Linear Regression and Logistic Regression. Both begin with a linear combination of predictors. The difference is that Logistic Regression applies a nonlinear transformation so that the final output stays in the interval from 0 to 1. That is why the model can be interpreted as a probability model rather than as a straight-line predictor.

The same model can be written on the odds or log-odds scale. The logarithm of the odds is called the logit. Writing the model in logit form makes the coefficient interpretation especially clear because each coefficient changes the log-odds linearly [4].

$$\text{odds}_i = \frac{p_i}{1 - p_i}$$

$$\log \left(\frac{p_i}{1 - p_i} \right) = z_i$$

A natural beginner question is why ordinary least squares is not used here. The reason is that a straight-line fit to binary outcomes is unsatisfactory in several ways. It can predict values outside the interval from 0 to 1, it does not model the mean-variance structure of binary data appropriately, and it does not reflect the nonlinear relation between the predictors and the probability scale. Logistic Regression solves this by combining a linear predictor with a nonlinear probability transformation [1,2,4].

Instead of least squares, Logistic Regression is commonly estimated by maximum likelihood estimation (MLE). The likelihood asks the following question: for a given choice of coefficients, how probable is the class pattern that we actually observed? A good set of coefficients should make the observed class labels likely under the model. The likelihood for independent binary observations is given below [4].

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

$$J(\boldsymbol{\beta}) = -\ell(\boldsymbol{\beta})$$

Because products of many probabilities can become cumbersome, the logarithm of the likelihood is usually maximised instead of the raw likelihood. Equivalently, one may minimise the negative log-likelihood, which is often called the loss or cost function. In introductory ML language, this is also closely related to cross-entropy loss. Unlike OLS in simple linear regression, Logistic Regression does not have a simple closed-form solution, so its coefficients are usually obtained by numerical optimisation methods [1,4].

Mathematically, Linear Regression and Logistic Regression therefore share an important common idea: both use a linear predictor. They differ in the output scale and in the estimation principle. Linear Regression predicts a continuous value and is usually estimated by least squares. Logistic Regression predicts a probability and is usually estimated by maximising likelihood. This comparison is central for beginners because it shows that the two models are related, but not interchangeable.

8 Model Interpretation and Evaluation for Logistic Regression

The first quantity to interpret in Logistic Regression is the predicted probability. If a model assigns a probability of 0.82 to machined-surface defect, it is saying that the observation is strongly associated with the distress class under the fitted model. It is not claiming certainty. Probability-based interpretation is especially valuable in engineering because it allows decisions to be linked to risk tolerance and consequence severity.

Coefficient interpretation is often easiest on the odds-ratio scale. If a coefficient for a predictor x_j equals β_j , then increasing x_j by one unit multiplies the odds by $\exp(\beta_j)$, provided the other predictors are held fixed. An odds ratio greater than 1 indicates increasing odds of the target class. An odds ratio below 1 indicates decreasing odds. This interpretation is mathematically exact and often more useful than trying to reason directly on the probability scale [4].

To convert probabilities into class labels, the modeller selects a threshold t . A common introductory rule is to classify as class 1 when the predicted probability is at least t , and as class 0 otherwise. Once this decision rule is applied, classification outcomes can be organised in a confusion matrix. The matrix compares predicted classes with actual classes using four counts: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).

$$\hat{y} = \begin{cases} 1, & \text{if } p \geq t \\ 0, & \text{if } p < t \end{cases}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

	Predicted negative	Predicted positive
Actual positive	FN False negative	TP True positive
Actual negative	TN True negative	FP False positive

Figure 5. Confusion matrix showing the four basic classification outcomes. Adapted from ThresholdTom (2019) [15].

Accuracy is the proportion of all cases classified correctly. Precision measures how many predicted positives are truly positive. Recall measures how many actual positives the model successfully identifies. Specificity, another useful measure, is the proportion of actual negatives correctly identified. In safety-related engineering classification, recall can be more important than accuracy, because missing a hazardous case may be much more serious than producing a false alarm.

These measures must be interpreted in context. Suppose a bearing-failure screening model has high accuracy because most components are safe, but it misses many of the few dangerous components. Such a model may be unacceptable despite apparently strong overall performance. This is why beginners should never rely on accuracy alone when classes are imbalanced or when the consequences of false negatives and false positives are unequal [1,4,10].

The key conceptual difference from Linear Regression should now be clear. Linear Regression fits a line or hyperplane to estimate a numerical response. Logistic Regression estimates the probability that an observation belongs to a class. The probability is then turned into a classification only after a

threshold is chosen. This extra probability stage is what makes Logistic Regression especially useful in risk, compliance, and condition-assessment problems.

Part summary

- Logistic Regression uses a linear predictor, but transforms it into a probability by the logistic function.
- Its coefficients act on the log-odds scale, and exponentiating a coefficient gives an odds ratio.
- Evaluation must consider the confusion matrix and the engineering consequences of false positives and false negatives, not just overall accuracy.

Part IV. Comparison, Applications, and Worked Examples

Bringing the two models together and connecting them to Mechanical Engineering decisions.

9 Comparing Linear Regression and Logistic Regression

Although Linear Regression and Logistic Regression answer different kinds of questions, they are closely related mathematically. Both models build a linear predictor from the inputs. Both use coefficients that quantify how predictors contribute to the model. Both are valued because their structure is interpretable. The crucial difference lies in what is being modelled and how the coefficients are estimated [1,2,4].

Table 6. Direct comparison of Linear Regression and Logistic Regression.

Feature	Linear Regression	Logistic Regression
Target variable	Continuous numerical quantity	Categorical outcome, often binary
Model output	Predicted numerical value	Predicted probability of class 1
Geometric picture	Fitted line or hyperpworkcell	Probability curve after logistic transformation
Estimation principle	Least squares	Maximum likelihood
Coefficient meaning	Change in predicted response per unit predictor change	Change in log-odds; exponentiated coefficient gives odds ratio
Main diagnostics	Residual plots, R-squared, error magnitudes	Confusion matrix, probability interpretation, precision, recall, accuracy
Common engineering question	How much?	Which class? or how likely?
Typical Mechanical Engineering use	Strength, deflection, deflection, material property prediction	Risk, compliance, distress occurrence, condition classification

Important distinction

Linear Regression fits the expected value of a continuous quantity. Logistic Regression fits the probability that an observation belongs to a class. A class label appears only after a probability is compared with a chosen threshold.

10 Mechanical Engineering Applications

The value of these models becomes much clearer when they are placed inside real engineering workflows. Mechanical Engineering datasets often contain measured inputs such as geometry, material composition, loading, environmental conditions, or condition indicators. The outputs may be numerical performance quantities or categorical condition states. Because Linear Regression and Logistic Regression are interpretable, they are especially useful when engineers must explain results to decision-makers, review authorities, contractors, or other designers [5].

Table 7. Illustrative Mechanical Engineering applications of Linear Regression and Logistic Regression.

Area	Engineering question	Typical predictors	Output	Model	Illustrative source
Materials	Predict mechanical material strength	Water-cement ratio, age, cement content, admixture content	MPa	Linear	[6]
Structures	Predict deflection or service deflection	Load level, geometry, stiffness, material properties	Continuous	Linear	[5]
Mechanics of materials	Classify site or material suitability	Plasticity, gradation, water content, density	Suitable / unsuitable	Logistic	[5]
Machined surfaces	Predict distress occurrence probability	Load, climate, surface properties, age	Yes / no	Logistic	[9]
Mechanical components	Classify condition or component rating category	Inspection findings, age, component type, environment	Ordered class	Logistic	[8]
Coolant quality	Predict regulatory compliance status	Real-time surrogate measurements, flow, season	Compliant / non-compliant	Logistic	[7]
Slopes	Predict failure likelihood	Load, alignment angle, lubrication, material properties	Failure / no failure	Logistic	[10]

Material strength prediction is a classic regression example. The output is continuous, so Linear Regression is a natural starting point. A model may relate strength to water-cement ratio, cement content, age, or admixture dosage. Even when more flexible models are later considered, a linear model often provides a useful baseline and a clear interpretation of directional effects [5,6].

Condition and risk classification problems are equally common. A vehicle systems team may wish to estimate the probability that machined-surface defect will occur during a maintenance cycle. A mechanics of materials and structures team may wish to estimate the probability that a slope enters a failure class under intense flow-rate. A water-quality monitoring programme may wish to classify whether a sample will exceed a threshold value using real-time surrogate measurements. These are exactly the kinds of questions for which Logistic Regression is especially well suited [7,9,10].

In structural condition assessment, the relevant outcome is sometimes ordered rather than merely binary. For example, mechanical component elements may be rated as good, fair, or poor. In such cases, an extension such as ordinal logistic regression can preserve the ordered nature of the categories while still retaining the interpretability of coefficient-based modelling [8].

Across all these applications, domain knowledge remains essential. The choice of predictors should reflect mechanics, materials science, site understanding, measurement quality, and the intended decision. Data-driven models do not replace engineering knowledge; they use data to complement it.

11 Worked Example: Linear Regression for Mechanical Material Strength

Teaching example

This is a simplified instructional example designed to illustrate model form and interpretation rather than full laboratory realism. The example is consistent with the well-known importance of mix variables in strength prediction, as discussed in the material-modelling literature [6].

Suppose a student wants to understand how water-cement ratio influences 28-day mechanical material strength. Consider the simplified dataset below, where x denotes water-cement ratio and y denotes measured 28-day compressive strength in megapascals (MPa).

Table 8. Simplified Linear Regression example for mechanical material strength.

Water-cement ratio x	Observed strength y (MPa)	Fitted strength $\hat{y}$ (MPa)	Residual $e = y - \hat{y}$ (MPa)
0.35	54.0	53.8	0.2
0.40	49.0	49.3	-0.3
0.45	45.0	44.8	0.2
0.50	40.0	40.3	-0.3
0.55	36.0	35.8	0.2

Fitting a least-squares line to these five observations gives the following model:

$$\hat{y} = 85.3 - 90x$$

The negative slope has a clear interpretation. An increase of 0.01 in water-cement ratio reduces the predicted 28-day compressive strength by about 0.9 MPa because the slope is -90 MPa per unit water-cement ratio. The intercept, 85.3 MPa, is the formal prediction when $x = 0$. Although that exact condition is not physically meaningful here, the intercept is still necessary to position the line.

If the water-cement ratio is 0.42, the model predicts a strength of $85.3 - 90(0.42) = 47.5$ MPa approximately. The residuals in Table 8 are very small, and the coefficient of determination is about 0.9985. For a teaching example, this indicates that the fitted line captures the observed trend extremely well.

The engineering interpretation is more important than the arithmetic. The model reflects the widely recognised inverse relation between water-cement ratio and strength: as water-cement ratio increases, strength tends to decrease. However, a real design model would ordinarily include more variables, such as curing age, cement content, aggregate characteristics, and admixtures. A one-predictor model is helpful for understanding the idea, but it is not a full mix-design tool [6].

This example also shows why residuals matter. The fitted values are close to the observations, but the residuals are not exactly zero. That is realistic. A regression model explains a pattern; it does not eliminate all variability.

12 Worked Example: Logistic Regression for Slope Failure Likelihood

Teaching example

This example is intentionally simplified so that the interpretation of odds, probabilities, and thresholds is transparent. It is inspired by the use of Logistic Regression in slope-failure studies such as [10].

Assume that R denotes 24-hour flow-rate in millimetres and that the response class is whether a bearing failure occurs. Consider the following Logistic Regression model:

$$\log\left(\frac{p}{1-p}\right) = -4 + 0.08R, \quad p = \frac{1}{1 + e^{-(-4+0.08R)}}$$

The first equation is the log-odds form. The second equation converts those log-odds into a probability p of failure. This model implies that higher flow-rate increases the chance of failure because the flow-rate coefficient is positive.

Table 9. Predicted probabilities from the simplified slope-failure Logistic Regression example.

Flow-rate R (mm)	Linear predictor z	Predicted probability p	Class at threshold 0.5
30	-1.6	0.168	No failure predicted
40	-0.8	0.310	No failure predicted
50	0.0	0.500	Boundary case
60	0.8	0.690	Failure predicted
70	1.6	0.832	Failure predicted

When $R = 50$ mm, the linear predictor equals zero, which gives $p = 0.5$. In other words, 50 mm is the point at which this particular model is exactly undecided under a 0.5 threshold. At 40 mm, the model gives p about 0.31, whereas at 65 mm it gives p about 0.77. This illustrates the gradual probabilistic nature of Logistic Regression: the risk increases continuously before any hard classification is made.

The coefficient 0.08 also has an odds-ratio interpretation. Increasing flow-rate by 1 mm multiplies the odds of failure by $\exp(0.08)$, which is about 1.083. Increasing flow-rate by 10 mm multiplies the odds by $\exp(0.8)$, which is about 2.23. Thus an extra 10 mm of flow-rate more than doubles the odds of failure in this simplified model.

The engineering meaning is again more important than the arithmetic. The model does not say that flow-rate alone determines failure. Real slope behaviour also depends on slope angle, lubrication and moisture conditions, material strength, coolant return, vegetation, and geological structure. The example simply shows how Logistic Regression links a measured predictor to a probability of belonging to a failure class [10].

It also shows why threshold choice matters. If the consequence of missing a dangerous slope is severe, engineers may classify high risk at a threshold below 0.5 in order to increase recall. Logistic Regression is therefore valuable not only because it classifies, but because it provides a probability that supports risk-informed decisions.

Part summary

- Linear and Logistic Regression share a linear predictor, but they answer different engineering questions and use different output scales.
- Mechanical Engineering applications range from material-property prediction to failure-risk, compliance, and condition classification.
- Worked examples show how coefficients become meaningful only when translated into units, probabilities, and engineering decisions.

Part V. Practical Judgment for Beginners

Recognising assumptions, strengths, limitations, and study habits that lead to sound use of regression models.

13 Assumptions, Strengths, and Limitations

Both Linear Regression and Logistic Regression are powerful introductory models precisely because they are simple and interpretable. Their simplicity, however, means that the modeller must think carefully about assumptions and data quality. These methods work best when the chosen predictors are meaningful, the sample represents the intended application setting, and the relation between predictors and outcome is not too far from the assumed model form [1-4].

13.1 Main assumptions of Linear Regression

The central practical assumption in Linear Regression is that the mean response is approximately linear in the predictors. If the true relation is strongly curved, a straight-line model may be systematically biased. Independence of errors is also important, especially in repeated measurements, spatial data, or time-indexed records where neighbouring observations may be related. Constant variance matters because changing residual spread can make some regions of the predictor space more uncertain than others. Approximate normality of errors is most important when one wants formal interval estimates or hypothesis tests, rather than merely a fitted line [3].

Linear Regression is also sensitive to outliers and influential observations. An outlier is an observation whose response differs unusually from the general pattern. An influential point is one whose presence can noticeably change the fitted coefficients. In engineering datasets, these may arise from measurement errors, transcription mistakes, rare field conditions, or genuinely extreme but important cases. Such observations should be investigated, not automatically discarded.

13.2 Practical assumptions of Logistic Regression

Logistic Regression assumes that observations are independent and that the log-odds of the target class change linearly with the predictors. Notice that this is not the same as assuming a linear relation on the probability scale. The probability curve is nonlinear by design. Another practical requirement is that the classes should both be represented in the data. If one class is extremely rare, the fitted model may become unstable or appear accurate while failing to identify important positive cases [4].

Logistic Regression can also suffer from multicollinearity when predictors are strongly redundant. In addition, some datasets exhibit complete separation, meaning that a predictor or combination of predictors perfectly separates the classes in the sample. In such cases, coefficient estimates can become extremely large and unstable. This is a reminder that a mathematically fitted model is not always a practically reliable one.

13.3 Strengths of both models

The main strengths of Linear Regression and Logistic Regression are simplicity, transparency, and interpretability. Coefficients have clear meanings. Assumptions can be checked. Model structure can be explained to non-specialists. These features make the models highly valuable in engineering environments where decisions must be justified clearly and where understanding the direction and magnitude of effects is often as important as obtaining a prediction [1,3,4].

These models are also foundational. Many later ML methods are easier to understand after a student has mastered the ideas of predictors, targets, residuals, likelihood, loss functions, and generalisation in

the context of Linear Regression and Logistic Regression. In that sense, these methods are not just useful tools; they are conceptual entry points into the wider field [1,2].

13.4 Limitations and caution points

The limitations follow directly from the strengths. Because the models are relatively simple, they may perform poorly when the true relation is highly nonlinear, when interactions between variables are strong but omitted, or when the data contain complex structures not captured by a linear predictor. Linear Regression can be misleading under strong curvature or heavy outlier influence. Logistic Regression can be misleading when class imbalance is severe or when the chosen threshold ignores the real cost of different errors.

Multicollinearity is another common limitation. This term means that two or more predictors contain nearly the same information. When that happens, the model may still predict reasonably, but the individual coefficients become unstable and difficult to interpret. Omitted-variable bias is also important: if a major explanatory factor is missing, the included coefficients may absorb effects that do not really belong to them.

Extrapolation deserves special caution. A model fitted within one data range should not automatically be trusted outside that range. A regression line that fits material strengths between certain mix proportions does not guarantee correct predictions far beyond those proportions. A Logistic Regression model fitted under one climate or material regime may not generalise reliably to a very different region. Engineering judgement must therefore accompany statistical results.

13.5 Brief conceptual comparison with other simple approaches

Other introductory modelling approaches exist. Decision trees, for example, can capture nonlinear patterns and interactions by splitting the data into regions, but they may be less stable and less smooth than regression-based models. Nearest-neighbour methods can adapt flexibly to local patterns, but they are often less interpretable and can depend strongly on the scale of the inputs. Linear Regression and Logistic Regression are usually taught first because they show the central logic of modelling in a mathematically transparent form [1,2].

Table 10. Strengths and limitations of the two regression-based models.

Aspect	Linear Regression	Logistic Regression
Main strength	Direct unit-based interpretation for continuous outputs	Natural probability-based interpretation for class membership
Works well when	The trend is approximately linear and residual behaviour is well behaved	The log-odds relation is approximately linear and classes are sufficiently represented
Common weakness	Sensitive to outliers, nonlinearity, and extrapolation	Sensitive to class imbalance, separation, and misspecified predictors
Interpretation benefit	Coefficients describe changes in the predicted quantity	Coefficients describe changes in log-odds and odds ratios
Typical beginner error	Using it for categorical outcomes	Treating the output as a direct certainty rather than a probability

14 Beginner Support and Study Advice

Students encountering ML for the first time often struggle not because the equations are impossible, but because the modelling questions are unfamiliar. The best strategy is to begin by identifying the target variable, the available predictors, the engineering decision that follows from the model output, and the kinds of error that would matter in practice. Once those are clear, the mathematics becomes far easier to place in context.

14.1 Common beginner mistakes

Table 11. Common beginner mistakes and better ways to think about them.

Common mistake	Better way to think
Calling every prediction problem “regression”	First inspect the output type. Continuous outputs suggest regression; categorical outputs suggest classification.
Thinking Logistic Regression predicts only labels	It predicts probabilities first, and labels only after a threshold is chosen.
Judging a model only by one number such as accuracy or R-squared	Use several diagnostics and ask whether the model makes engineering sense.
Ignoring units and physical meaning	Interpret coefficients in engineering units and check whether signs and magnitudes are plausible.
Assuming a good fit on old data guarantees future reliability	Always ask how well the model will generalise to new cases and whether the data are representative.

14.2 Training, testing, overfitting, underfitting, and generalisation

In supervised learning, the training set is the part of the data used to estimate the model coefficients. The test set is a separate part of the data reserved for evaluating performance on previously unseen observations. This distinction is essential because a model must eventually be used on new cases, not merely on the data from which it was built [1,2].

Overfitting occurs when a model adapts too closely to the peculiarities or noise in the training data and therefore performs worse on new data. Underfitting occurs when a model is too simple to capture even the main structure of the data. Generalisation refers to the ability of a fitted model to perform well on new observations drawn from the same underlying setting. Good modelling seeks a balance: enough complexity to capture the important structure, but not so much complexity that the model memorises accidents of the training sample [1,2].

A practical decision rule for students

If the target is a numerical quantity such as deflection, strength, or deflection, begin by thinking “regression.” If the target is a label such as failure or no failure, compliant or non-compliant, or suitable or unsuitable, begin by thinking “classification.” Then ask what errors matter most in the engineering decision.

14.3 How Mechanical Engineering students should decide between regression and classification

A useful checklist is to ask four questions. First, what is the exact quantity or class that needs to be predicted? Second, is that quantity continuous or categorical? Third, what action will be taken after the prediction is made? Fourth, which mistakes are most costly or dangerous? If the model output will guide safety, compliance, or screening decisions, probability-based classification may be more informative than a single yes-or-no label.

Mechanical Engineering problems often begin with physical understanding and then move toward data-supported estimation. Students should therefore connect the modelling task back to the engineering purpose. If the problem asks “how much deflection is expected?”, the answer should be a continuous estimate. If the problem asks “is this site likely to fail under the forecast flow-rate?”, the answer should be a probability or a class decision. That simple contrast is one of the most important habits a beginner can develop.

14.4 Why these methods are usually taught first

Linear Regression and Logistic Regression are usually taught first because they combine conceptual clarity with mathematical depth. They introduce key ideas such as fitting, error, loss, probability, residuals, coefficients, and generalisation without hiding the structure behind a black box. They are simple enough to understand from first principles, yet rich enough to support serious engineering interpretation. For that reason, mastery of these two models provides a strong basis for later study in both statistics and ML [1,2,4].

Part summary

- Beginners should always identify the target variable, the engineering decision, and the cost of different errors before choosing a model.
- Training and testing address generalisation: a good model must work on new data, not only on the data used for fitting.
- These methods are taught first because they are interpretable, mathematically transparent, and foundational for later study.

15 Conclusion

Linear Regression and Logistic Regression occupy a special place in introductory AI and ML because they connect mathematical rigour with practical interpretability. Linear Regression models continuous quantities by fitting a line or hyperplane that minimises squared errors. Logistic Regression models class membership by transforming a linear predictor into a probability and then, if needed, applying a decision threshold [1-4].

For Mechanical Engineering students, this distinction is more than a technical detail. It separates questions about magnitude from questions about category and risk. Material strength, deflection, and deflection are naturally continuous and therefore suit Linear Regression. Material suitability, machined-surface defect occurrence, structural condition states, water-quality compliance, and slope-failure likelihood are naturally classification problems and therefore suit Logistic Regression or its extensions [5-10].

These models are not universal solutions, but they are exceptionally valuable starting points. They teach how to think about predictors and responses, how to connect equations to engineering meaning, how to examine model assumptions, and how to interpret outputs responsibly. Students who

understand these two models well have a strong conceptual and mathematical basis for further study in statistical learning, risk assessment, and data-informed Mechanical Engineering decision-making.

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